Total No. of Questions : 8]

SEAT No. :

PC2455

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B.E. (Information Technology)

DISTRIBUTED SYSTEMS

(2019 Pattern) (Semester - VIII) (414450)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Figures to right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Why is clock synchronization using a global time impossible in a distributed system? Illustrate the active time server based centralized clock synchronization algorithm. **[9]**

b) What is the purpose of Message Passing Interface? Illustrate the architectural model for MPI using send and receive primitives. **[8]**

OR

Q2) a) Discuss the purpose of overlay network. Describe in brief, any three types of Overlay. **[9]**

b) What are the two important properties of token-based approach? Explain token-ring algorithm to achieve mutual exclusion in a distributed system. **[8]**

Q3) a) Describe the following independent axes for defining inconsistencies with examples: **[9]**

i) Deviation in numerical values between replicas.

ii) Deviation in staleness between replicas.

iii) Deviation with respect to ordering of update operations.

b) With a neat diagram, explain the three types of replicas and their logical organization. **[9]**

OR

P.T.O.

Q4) a) What is checkpointing in a distributed system? Explain the working of Coordinated checkpointing recovery mechanism. **[9]**

b) Discuss and compare "Push versus Pull Protocols" propagation design issue for content distribution. **[9]**

Q5) a) What is a Directory Service? What is the difference between DNS and x500? Describe in detail the components of X.500 service architecture. **[8]**

b) What are the key design issues for distributed file systems? Describe the requirements for distributed file systems. **[9]**

OR

Q6) a) Explain in brief, the two places of client-side web caching? Explain cooperative caching with suitable diagram. **[8]**

b) What are web services? Describe with a suitable diagram the general organization of the Apache web server. **[9]**

Q7) a) Explain the following in brief: **[9]**

i) Wearable devices

ii) PVM

iii) JINI

b) Explain in brief, the key features of Zabbix including auto-discovery, triggers, or dashboards. **[9]**

OR

Q8) a) Explain in brief, the following Distributed System monitoring tools: **[9]**

i) Nagios

ii) Zabbix

b) Provide an overview of Mach and CHORUS microkernels. How are memory management techniques used to avoid physical copying of data in Mach and CHORUS? **[9]**

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